



CATHOLIC UNIVERSITY OF EASTERN AFRICA

**PROJECT PROPOSAL FOR FINAL YEAR STUDY IN
SCI 400 RESEARCH PROJECT**

BY

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PROJECT TITLE

AI-Powered Automated Investment Portfolio Analyzer

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CHAPTER ONE: INTRODUCTION

1.1 Background

Financial planning and investment analysis are important for both individuals and organizations seeking to achieve long-term financial goals. In many developing economies such as Kenya, a large portion of the population lacks access to professional financial advice due to the high cost of consultancy services and limited awareness of digital investment tools. According to the Central Bank of Kenya (CBK, 2024), over 60% of working adults manage their finances manually or through basic spreadsheets. These methods are prone to human error, data loss, and inconsistent decision-making.

Globally, tools like Yahoo Finance and Mint provide access to market data and personal finance tracking. However, these tools often lack localization for African markets and do not offer personalized, AI-driven investment advice. Traditional spreadsheets such as Microsoft Excel remain common but have an error rate of approximately 30% (NetSuite, 2024). This makes it difficult for users to make data-driven decisions in fast-changing economic environments.

In Kenya, the growing use of mobile banking and digital investment apps such as Safaricom's M-Pesa, Absa Timiza, and NCBA Loop has created a foundation for financial inclusion. However, these applications primarily focus on transactions and savings rather than intelligent, goal-based investment planning. The lack of advanced analytical tools for local investors results in poor financial forecasting, limited portfolio optimization, and missed investment opportunities.

Emerging technologies such as Artificial Intelligence (AI), Machine Learning (ML), Vector Databases, and Large Language Models (LLMs) now make it possible to develop automated systems that provide personalized, real-time financial insights. These technologies can analyze large datasets, simulate future scenarios, and generate user-specific recommendations using natural language. Therefore, this research focuses on developing an Automated Investment Portfolio Analyzer that leverages AI, RAG (Retrieval-Augmented Generation), and interactive dashboards to deliver affordable, data-driven investment advice in Kenya.

1.2 Problem Statement

Many Kenyans lack access to affordable, personalized financial planning tools. Professional financial advisors charge between KES 10,000 to 60,000 per session, making them inaccessible to most middle-income earners and small businesses. Existing digital solutions such as Yahoo

Finance and Excel provide general insights but lack the ability to generate personalized investment plans or forecast future outcomes based on user goals.

According to the 2025 Association for Financial Professionals (AFP) report, nearly 60% of individuals fail to make informed financial decisions due to the absence of automated financial analysis tools. Furthermore, over 70% of Kenyan small business owners rely on manual bookkeeping and spreadsheets, which are slow, error-prone, and difficult to scale (Kenya Institute for Public Policy Research and Analysis [KIPPRA], 2024).

The lack of real-time analytics and interactive investment dashboards prevents users from understanding risks, visualizing returns, or adjusting strategies dynamically. This study, therefore, aims to develop a web-based AI-powered investment portfolio analyzer that integrates Yahoo Finance data, vector databases, LLMs, and interactive dashboards. The system will provide predictive analytics, portfolio simulations, and personalized recommendations, helping users make informed financial decisions faster and more accurately.

1.3 Objectives of the Study

1.3.1 General Objective

To design and develop an AI-driven Automated Investment Portfolio Analyzer that provides personalized, affordable, data-based investment recommendations for Kenyan users.

1.3.2 Specific Objectives

1. To integrate Yahoo Finance data and Kenyan financial datasets into a secure, scalable database for analytics.
2. To review and implement Machine Learning models for forecasting and simulation.
3. To develop a chatbot and ChatPDF assistant that provide conversational, document-based financial insights.
4. To evaluate system performance in terms of accuracy, usability, and effectiveness in supporting financial decision-making.
5. To design and develop an interactive financial dashboard that visualizes investment data in real time.

1.4 Justification of the Study

The study is justified by the growing need for financial literacy, automation, and accessibility in Kenya's financial ecosystem. According to the Financial Sector Deepening (FSD) Kenya Report

(2024), nearly half of Kenyans lack reliable access to financial advisory services, yet most own smartphones capable of supporting intelligent mobile apps.

Existing research (World Bank, 2023; Deloitte, 2024) emphasizes the potential of AI-driven financial tools to democratize investment guidance. However, most global systems are tailored to Western markets and do not consider local challenges such as irregular income streams, inflation fluctuations, and limited digital advisory coverage in Kenya.

Developing this system contributes to both academic and practical knowledge by demonstrating how Machine Learning, LLMs, and Vector Databases can automate financial planning in emerging economies. It also aligns with Kenya's Digital Economy Blueprint (2019), which encourages the integration of advanced data analytics to enhance economic decision-making. This study's outcome will benefit:

1. Individuals, by offering low-cost, personalized investment guidance.
2. Small businesses, by improving data-driven budgeting and forecasting.
3. Researchers, by contributing to the growing literature on AI in financial planning within the African context.

1.5 Scope of the Study

This research focuses on developing an Automated Investment Portfolio Analyzer targeting individual investors and small businesses in Kenya. The project will integrate Yahoo Finance data, Kenyan market data, and user-entered financial goals into a centralized database. Key features include:

- i. Real-time dashboards displaying investment performance.
- ii. AI-driven personalized portfolio recommendations.
- iii. Chatbot for goal-based financial queries.
- iv. ChatPDF assistant for document-based insights.
- v. Notification system (via MCP) for market alerts.
- vi. Vector Database and RAG integration for semantic search and intelligent responses.

The prototype will be implemented using Python, Hadoop (for big data handling), FAISS or Pinecone (for Vector DB), LangChain (for LLM orchestration), and Streamlit or Dash (for dashboard visualization). Testing will be limited to simulated and real-time Yahoo Finance datasets and a sample group of Kenyan users.

1.6 Organization of the Report

The research is organized into six chapters as follows:

- i. **Chapter One:** Introduces the study, stating the background, problem, objectives, justification, scope, and organization of the research.
- ii. **Chapter Two:** Reviews related literature, existing systems, emerging trends, and research gaps in AI-based financial planning.
- iii. **Chapter Three:** Describes the research methodology, including system analysis, design, data collection, and implementation approaches.
- iv. **Chapter Four:** Provides system analysis, including feasibility study, functional requirements, and Data Flow Diagrams.
- v. **Chapter Five:** Focuses on system design, database modeling, and interface mock-ups.
- vi. **Chapter Six:** Presents the system implementation, testing, and validation results, followed by the final summary and recommendations.

CHAPTER 2: LITERATURE REVIEW

2.1 Chapter Introduction and Methodology of Literature Review

This chapter reviews existing studies, systems, and technological solutions related to automated investment management, financial advisory systems, and artificial intelligence integration in the finance sector. The goal is to analyze previous work globally and locally to understand how automated portfolio systems have evolved and to identify the gaps that this study seeks to address within the Kenyan investment context.

The literature review followed a systematic research approach using secondary sources such as academic journals, online financial reports, and case studies. The sources were retrieved from platforms such as Google Scholar, ScienceDirect, IEEE Xplore, and the Central Bank of Kenya (CBK) reports. Each selected study was analyzed based on its relevance to automated investment platforms, financial technology (FinTech) development, artificial intelligence adoption, and local accessibility in Kenya.

2.2 History of the Research Topic

The history of automated investment management dates back to the early 2000s with the rise of robo-advisors, which used algorithmic models to provide digital financial advice. Globally, companies such as Betterment and Wealthfront were among the pioneers, offering automated portfolio management based on user goals and risk tolerance. These platforms significantly

reduced advisory costs compared to traditional human financial advisors, who typically charge \$100–\$500 per session (Statista, 2024).

In Kenya, the investment sector has experienced steady growth but remains limited by low financial literacy, manual investment tracking, and high advisory costs (Capital Markets Authority, 2023). Platforms such as Safaricom’s M-Akiba, Chumz, and Absa’s Timiza have introduced digital investment tools, but most focus on micro-savings rather than personalized portfolio analytics. The lack of integrated AI-driven tools for goal-based investment recommendations continues to hinder efficient decision-making among Kenyan investors, particularly young adults and small business owners.

2.3 Review of Related Prototypes and Systems

This section reviews five relevant systems and studies, both global and local, that apply automation, artificial intelligence, and financial analytics.

2.3.1 Global Case 1: Betterment (United States)

Betterment is one of the leading robo-advisory platforms globally. It uses automated algorithms to build diversified portfolios based on the client’s financial goals, risk levels, and investment horizons. The system relies on data analytics, machine learning, and predictive modeling to rebalance portfolios automatically. While it offers excellent automation, its main limitation is limited adaptability to emerging markets like Kenya, where data availability and investment structures differ (Betterment, 2024).

2.3.2 Global Case 2: Wealthfront (United States)

Wealthfront employs Monte Carlo simulations to estimate the probability of achieving investment goals. It integrates APIs for real-time market data and uses AI to generate financial insights. The platform is powerful but depends heavily on U.S.-based market data, making it unsuitable for developing economies. Its technology base, however, provides a framework for developing similar tools locally using Kenyan market data (Wealthfront, 2024).

2.3.3 Regional Case 3: Acorn Investment Platform (South Africa)

Acorn allows users to invest spare change into selected portfolios using automation and mobile integration. It demonstrates how mobile-first design and simple interfaces can increase financial inclusion. However, it lacks deep learning capabilities or real-time predictive analytics, features that could enhance returns and risk evaluation for investors in dynamic markets like Kenya (African FinTech Report, 2024).

2.3.4 Local Case 4: Chumz App (Kenya)

Chumz enables Kenyans to set savings goals and invest small amounts directly through their mobile phones. The app integrates behavioral economics principles to encourage saving. Despite its popularity, it lacks AI personalization and interactive investment dashboards. The proposed system Automated Investment Portfolio Analyzer will extend this idea by incorporating machine learning, vector databases, and conversational AI for personalized investment recommendations (TechTrends Kenya, 2025).

2.3.5 Local Case 5: M-Akiba (Kenya)

M-Akiba, developed by the Government of Kenya, allows citizens to invest in government bonds via mobile platforms. It has improved access to investment opportunities but does not offer data visualization, AI-driven advisory, or interactive reporting. Therefore, while it increases accessibility, it does not assist users in making informed decisions about diversified portfolios (Capital Markets Authority, 2024).

Table 1: Summary of Reviewed Systems

System	Location	Technology Used	Strengths	Limitations
Betterment	USA	AI, Data Analytics	Automated rebalancing	Not localized for Africa
Wealthfront	USA	ML, Monte Carlo Simulation	Predictive analytics	High-cost entry, foreign data
Acorn	South Africa	Mobile Integration	Encourages saving culture	No predictive modeling
Chumz	Kenya	Mobile app, API	Simple savings automation	No AI or dashboard analytics
M-Akiba	Kenya	Mobile & Telecom Integration	Promotes inclusion	No personalized advisory

2.4 Emerging Trends and Patterns in the Research Area

Recent research shows a global shift toward AI-integrated financial systems that combine machine learning, natural language processing (NLP), and automation to personalize investment experiences. The rise of Large Language Models (LLMs) like GPT and Retrieval-Augmented

Generation (RAG) has enabled chatbots to offer context-aware investment advice and interactive report generation.

In Kenya, the FinTech revolution has accelerated with platforms such as Safaricom's M-Pesa, Absa Timiza, and NCBA Loop, yet most focus on digital banking and lending rather than investment analytics. There is growing potential for AI-driven investment systems that offer:

- i. Interactive dashboards for performance tracking.
- ii. Predictive modeling of market risks using Monte Carlo simulations.
- iii. Chatbots and ChatPDF systems for document-based learning.
- iv. Vector databases for semantic data retrieval in financial queries.
- v. Real-time notification systems for investment updates.

These trends highlight a clear opportunity to merge data science, finance, and AI to bridge Kenya's financial advisory gap.

2.5 Research Gap

From the reviewed systems, it is evident that while global platforms like Betterment and Wealthfront provide advanced automation, they do not fit Kenya's unique market conditions. Local tools like Chumz and M-Akiba increase accessibility but lack AI-driven personalization, real-time analytics, and intelligent reporting.

Hence, the research gap lies in the absence of an integrated, AI-powered system tailored for Kenya's investment environment. The Automated Investment Portfolio Analyzer will fill this gap by developing a web-based intelligent system that combines:

- i. Databases for data storage and management,
- ii. Interactive dashboards and reports,
- iii. Vector databases for semantic search,
- iv. Chatbots and LLMs for personalized financial conversations, and
- v. Real-time notifications for market insights.

This solution will make personalized investment planning affordable, accessible, and data-driven, helping ordinary Kenyans make informed financial decisions.

2.6 Chapter Summary

This chapter reviewed global and local systems addressing automated investment management. It revealed that while global solutions are technologically advanced, they are not localized for Kenyan markets. Conversely, local platforms promote financial inclusion but lack advanced

analytics and personalization. The emerging trend points toward the integration of AI, LLMs, and data-driven decision support systems to enhance investment advisory. The identified research gap justifies the need for an AI-based Automated Investment Portfolio Analyzer focused on Kenya's investment ecosystem.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Chapter Introduction

This chapter explains the methods and procedures used to design, develop, and test the AI-Powered Automated Investment Portfolio Analyzer, which aims to automate and personalize investment planning and decision-making for Kenyan investors.

The methodology outlines the stages of requirement determination, system analysis, design, development, and testing, which were guided by System Development Life Cycle (SDLC). The methodology follows a system development life cycle (SDLC) model combining Agile and Prototyping approaches, which allow for iterative refinement, stakeholder feedback, and adaptive design suited for AI-driven applications.

3.2 Methodology of Requirement Determination

The requirements determination process involved collecting, defining, and prioritizing user needs from both investors and financial analysts.

3.2.1 Functional and Non-Functional Requirements

Requirements were defined through both functional and non-functional perspectives.

- i. **Functional requirements** include user registration, portfolio creation, chatbot and ChatPDF integration, personalized investment planning, and performance report generation.
- ii. **Non-functional requirements** include scalability, system reliability, security, scalability, accuracy, and real-time data updates.

3.2.2 Data Collection Techniques

To gather relevant information on how Kenyan investors interact with financial systems, the following methods were used:

- i. **Interviews:** Conducted with investment officers and financial analysts from banks such as Absa, NCBA, and Cooperative Bank to understand current digital investment challenges.
- ii. **Questionnaires:** Distributed to 10 individual investors and 10 SMEs in Nairobi to identify user needs such as affordability, simplicity, and automation preferences.

- iii. **Document Review:** Analysis of reports from the Capital Markets Authority (CMA, 2024) and Kenya Bankers Association (2025) to gather insights on financial technology trends.
- iv. **Observation:** Study of how users interact with current online investment dashboards (e.g., Absa Wealth and Safaricom Mali) to identify UX/UI limitations.

Table 2: Data Collection Methods

Method	Target Group	Purpose	Outcome
Interviews	Investment Analysts, Bank IT Experts	Understand workflow and pain points	Identified need for automation and real-time dashboards
Questionnaires	Retail Investors, SMEs	Identify user preferences	Found high interest in simple AI-guided investment tools
Document Review	CMA Reports, Bank Publications	Analyze current tech usage	Confirmed gaps in personalized advisory systems
Observation	Web/Mobile Investment Portals	Assess usability	Identified lack of interactivity and real-time updates

3.3 Methodology for System Analysis

3.3.1 Current System Analysis

Kenyan investment banking still relies heavily on manual data processing, spreadsheets, and static reports. These systems are slow, prone to human error, and lack integration with real-time financial data sources. Investors and analysts spend up to 60% of their time on repetitive data collection and reconciliation tasks (NetSuite, 2024).

The current systems are characterized by:

- i. Manual financial data error-prone entry.
- ii. No personalized intelligent portfolio recommendations.
- iii. Static reports without predictive analysis.
- iv. No real-time alerts or notifications.

3.3.2 Data Flow Diagrams (DFDs)

- i. **Level 0 Context Diagram:** Investor inputs goals and data → LLM processes request → ML model analyzes risk → dashboard displays recommendation.
- ii. **Level 1 DFD:** Breaks system into modules (user management, data processing, ML model, reporting).

- i. **Level 2 DFD:** Highlights subprocesses like risk assessment, portfolio balancing, and reporting.

3.3.3 Normalization

The database normalization process will progress up to the Third Normal Form (3NF) to eliminate redundancy and maintain data integrity. For example:

- i. **First Normal Form (1NF):** Store investor details, portfolio records, and transaction data in separate tables.
- ii. **Second Normal Form (2NF):** Remove partial dependencies between portfolio and investor tables.
- iii. **Third Normal Form (3NF):** Ensure all non-key attributes depend only on the primary key.

3.4 Methodology for System Design

3.4.1 Proposed System Overview

The proposed system replaces manual reporting with an AI-powered web application. Key components include:

- i. **Frontend (Client Interface):** Interactive dashboards and reports built using React.js, Tailwind CSS, and Chart.js.
- ii. **Backend (Server Logic):** Implemented using Python Flask/Django, handling data processing, authentication, and AI interactions.
- iii. **Database Layer:** Relational Database (PostgreSQL) for transactional and portfolio data. Vector Database (FAISS or Pinecone) for semantic search and document embeddings.
- iv. **AI Layer:** Ollama Models and LLMs for conversational financial advice. AI Agents for automated decision-making and report summarization. ChatPDF for reading investment documents and policy files.
- v. **Notification MCP:** Sends user alerts for portfolio performance or market shifts.

Figure 3.2: Proposed System Architecture (Author, 2025)

(Suggested diagram: Multi-layered architecture showing user → chatbot/LLM → backend processing → vector DB → dashboards.)

3.4.3 Flowcharts

A flowchart will be used to represent the logical flow of tasks such as:

- i. User submits investment goal.
- ii. System fetches data from Yahoo Finance API.
- iii. LLM analyzes data.
- iv. Vector DB retrieves historical performance.
- v. Dashboard displays report.

3.4.4 User Interface (UI/UX) Design

Prototype interfaces were developed using Figma and Canva to simulate the user experience. The interface emphasizes clarity, simplicity, and financial data visualization.

Figure 3.3: Sample Dashboard Prototype (Author, 2025)
(Screenshot suggestion: Portfolio summary dashboard with charts and metrics)

3.4.5 Database Design (ERD)

The Entity Relationship Diagram (ERD) defines the key relationships between investors, portfolios, transactions, and performance metrics.

Figure 3.4: ERD for Proposed System (Author, 2025)

3.5 Methodology for System Development

The system development focuses on building a functional prototype (not deployment). It transforms the design into a working model through iterative coding and testing cycles.

3.5.1 Development Model

The Agile methodology was adopted to enable short development sprints, regular testing, and user-centered feedback. Each sprint focused on developing a key module such as the chatbot or dashboard.

3.5.2 Tools and Technologies

Table 3: Tools and Technologies

Component	Technology Used	Purpose
Frontend	React.js, Tailwind CSS	Build responsive dashboards
Backend	Python Flask/Django	Handle APIs and logic
Database	PostgreSQL, FAISS Vector DB	Store structured and semantic data

AI Models	Ollama LLMs	Provide financial reasoning
Chatbot	LangChain + RAG	Enable natural interaction
Data Source	Yahoo Finance API	Real-time stock and market data

3.5.3 Coding Standards

- i. Followed PEP8 for Python and ESLint for JavaScript.
- ii. Used environment variables (.env) for secret keys.
- iii. Applied modular programming to ensure code reusability.

3.6 Methodology for System Testing

Testing ensures the prototype meets the specified requirements and performs as expected.

3.6.1 Testing Techniques

- i. **Unit Testing:** Testing individual modules such as chatbot responses, database queries, and report generation.
- ii. **Integration Testing:** Ensuring components like vector DB and AI agents work together smoothly.
- iii. **User Acceptance Testing (UAT):** Involving real investors and analysts from Kenyan banks.
- iv. **Performance Testing:** Measuring response times and load under heavy traffic.

3.6.2 Testing Tools

- i. Postman for API endpoint testing.
- ii. Pytest and Selenium for automated test scripts.
- iii. JMeter for performance load testing.

3.7 Chapter Summary

This chapter detailed the methodology guiding the design, development, and testing of the AI-Powered Automated Investment Portfolio Analyzer. The approach combined Agile and Prototyping models to ensure iterative refinement and user involvement. Requirements were gathered from real investors and banking professionals. The analysis identified weaknesses in manual investment systems, while the design introduced AI-driven architecture with interactive dashboards and vector databases. Development used Python, React, and Ollama-based AI components, and testing verified functionality, integration, and user satisfaction.

CHAPTER FOUR: SYSTEM ANALYSIS

4.1 Chapter Introduction

This chapter presents the analysis of the existing investment management systems in Kenya and describes the methods used to identify gaps, user needs, and system requirements. It also discusses the feasibility study, functional and non-functional requirements, and the decomposition of the proposed system using relevant system analysis tools.

The aim of the analysis is to design a smart, AI-powered investment portfolio platform that can improve investment decision-making and performance tracking for Kenyan investors.

4.2 Description of the Current System

The current investment management process in most Kenyan financial institutions relies heavily on manual data entry, Excel-based reports, and static dashboards. Financial analysts and clients often download data from multiple sources such as Yahoo Finance, NSE (Nairobi Securities Exchange), and CMA (Capital Markets Authority) websites, then manually analyze it using spreadsheets or local applications.

Strengths of the Current System:

- i. It is easy to use since Excel and PDF reports are familiar to most users.
- ii. It requires minimal technical skills to generate summary reports.
- iii. It works well for small investment portfolios with limited transactions.

Weaknesses of the Current System:

- i. Data processing is slow and prone to human error.
- ii. There is no real-time data integration or AI-based forecasting.
- iii. The system cannot provide interactive visualizations or automated insights.
- iv. Users lack personalized recommendations based on risk profiles or market changes.
- v. Data from different sources is not centralized, making portfolio tracking difficult.

This analysis shows that there is a need for a modern, AI-driven investment management system that can automate data collection, analysis, and reporting in real-time.

4.3 Feasibility Study

A feasibility study was conducted to determine whether developing the AI-powered investment platform was practical and beneficial for Kenyan financial institutions and investors.

- i. **Technical Feasibility:** The system was found to be technically feasible since open-source tools such as Python, Django, PostgreSQL, LangChain, and Ollama models can be used to

implement the backend. Front-end frameworks such as React and TailwindCSS support the development of interactive dashboards. The use of vector databases and APIs also allows smooth integration with external market data sources.

- ii. **Economic Feasibility:** The project is cost-effective because most tools used are open-source. The system also reduces long-term operational costs by automating data analysis and reporting, which would otherwise require manual labor.
- iii. **Operational Feasibility:** The system is operationally feasible because users such as financial analysts and investment managers can easily adapt to the new platform after short training sessions. Its dashboards and chatbot interfaces are user-friendly and designed to improve workflow efficiency.
- iv. **Schedule and Budget Feasibility:** A project schedule was developed to guide the activities for the design and development of the AI-Powered Automated Investment Portfolio Analyzer. The project will run for 12 weeks and will focus only on analysis, design, and development phases. All tools and resources used are open-source or free academic licenses, meaning the project operates on a zero (0) budget.

Table 4: Project Schedule (Gantt Summary)

Phase	Activities	Duration (Weeks)	Deliverables
1	Requirements Determination	2	Requirement Specification Document
2	System Analysis	2	Feasibility Report, DFDs, and Context Diagram
3	System Design	3	ERD, UI Mockups, and Architecture Diagram
4	System Development (Prototype Only)	4	Functional Prototype Modules
5	Testing and Review (Prototype Testing)	1	Test and Evaluation Report
Total Duration		12 Weeks	

Table 5: Project Budget

Item	Description	Estimated Cost (KES)
Hardware	Personal laptop used for development	0
Software	Free and open-source tools (Python, React, Figma, Canva, PostgreSQL)	0
Data Collection	Conducted through online interviews and secondary data sources	0
Cloud/Hosting	Local simulation using localhost server for prototype testing	0
Miscellaneous	Academic research references and documentation	0
Total		0 KES

4.4 Requirements Analysis

Requirements analysis was carried out to identify both the functional and non-functional requirements of the proposed system.

Functional Requirements:

- i. The system allows users to create and manage investment portfolios.
- ii. The system retrieves and stores market data from multiple financial sources.
- iii. The system generates interactive reports and visual dashboards.
- iv. The chatbot and AI agent provide personalized investment insights.
- v. The system sends notifications via email or SMS using MCP integrations.
- vi. The system enables secure login and user authentication.
- vii. The system allows export of data in formats such as PDF, Excel, and CSV.

Non-Functional Requirements:

- i. Performance: The system responds to user queries in under three seconds.
- ii. Scalability: It handles multiple users and large datasets efficiently.
- iii. Security: Data is encrypted and protected through secure login protocols.
- iv. Usability: The interface is simple, intuitive, and accessible via web and mobile devices.
- v. Reliability: The system maintains data integrity and minimizes downtime.

4.5 Determining the Appropriate System Analysis Approach

Before system modeling, a comparison was made between Structured System Analysis and Design (SSAD) and Object-Oriented System Analysis and Design (OOSAD) to determine the most suitable approach for the project.

- i. **Structured System Analysis and Design (SSAD):** Focuses on processes, data flows, and logical decomposition of functions. Uses tools such as Context Diagrams, Data Flow Diagrams (DFDs), and Entity Relationship Diagrams (ERDs). Best suited for systems that are primarily data-driven and process-oriented.
- ii. **Object-Oriented System Analysis and Design (OOSAD):** Focuses on objects, classes, and their interactions. Uses tools such as Use Case Diagrams, Class Diagrams, and Sequence Diagrams. Best suited for large-scale systems emphasizing code reusability and inheritance.

Justification for the Chosen Approach: The Structured System Analysis and Design (SSAD) approach was selected for this project because:

- i. The system is data-driven, focusing on input, processing, and output of financial and investment data.
- ii. It emphasizes process flow (e.g., portfolio data collection, risk analysis, and report generation) rather than complex object relationships.
- iii. SSAD supports clear documentation, which is vital for early research-based projects.
- iv. It provides a strong foundation for future transition into Object-Oriented Design if the project proceeds to full implementation.

4.6 System Analysis and Modeling

Structured System Analysis and Design (SSAD) techniques were used to break down the proposed system into components. The main tools used were the Context Diagram and Data Flow Diagrams (DFD).

4.6.1 Context Diagram

The context diagram (Figure 4.1) illustrated the system as a single process interacting with external entities such as investors, administrators, and data sources.

(Figure 4.1: Context Diagram of the Proposed Investment Management System)

4.6.2 Level 1 DFD

The Level 1 DFD broke down the main system into subsystems such as:

- i. User Management
- ii. Market Data Processing
- iii. Portfolio Analysis
- iv. Report Generation
- v. AI Chatbot Interaction

(Figure 4.2: Level 1 DFD of the Proposed System)

4.6.3 Level 2 DFD

The Level 2 DFD further detailed how data flowed between modules such as the vector database, API layer, and dashboard interface.

(Figure 4.3: Level 2 DFD of the Proposed System)

4.7 Identification of System Users, Inputs, and Outputs

System Users:

- i. **Administrators:** Manage system settings, database, and user accounts.
- ii. **Financial Analysts:** Use the system for report generation and analysis.
- iii. **Investors/Clients:** Use dashboards and chatbots to track investments.

Inputs:

- i. Market data (e.g., stock prices, currency rates)
- ii. User information and portfolio details
- iii. External documents (PDFs, reports, etc.)

Outputs:

- i. Interactive dashboards and charts
- ii. Investment performance reports
- iii. AI-generated insights and alerts
- iv. Email and SMS notifications

4.8 Database Entities and Normalization

The database is designed to store structured financial data efficiently. Entities such as User, Portfolio, Investment, Transaction, and MarketData were identified.

Normalization was carried out up to the Third Normal Form (3NF) to eliminate redundancy and ensure data integrity.

Table 6: Example of Normalized Tables up to 3NF

Entity	Attributes	Description
User	User_ID, Name, Email, Password	Stores user details
Portfolio	Portfolio_ID, User_ID, Portfolio_Name	Links portfolios to users
Investment	Investment_ID, Portfolio_ID, Type, Amount	Tracks investments
MarketData	Market_ID, Date, Stock_Symbol, Price	Stores real-time data
Transaction	Transaction_ID, Investment_ID, Date, Type, Amount	Records transactions

4.9 Chapter Summary

This chapter presented the system analysis for the AI-powered investment portfolio management platform. The weaknesses of the current system were identified and analyzed, and a detailed feasibility study was conducted to ensure practicality. The functional and non-functional requirements were outlined, and the system was decomposed into logical components using Structured System Analysis and Design (SSAD) tools such as Context and DFD diagrams. Entities were identified and normalized up to the third normal form to reduce redundancy and improve efficiency. The decision to use SSAD was justified as the most suitable approach for this data-driven, research-level project that concludes at system development.

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